

KANGAN-AVSS: Kolmogorov-Arnold Network Based Generative Adversarial Networks for Audio-Visual Speech Synthesis

Subhayu Ghosh¹, Swapnil Saha², Nanda Dulal Jana¹

¹Department of Computer Science and Engineering, National Institute of Technology Durgapur, India.

²Department of Power Engineering, Jadavpur University, Kolkata, India.

Email: subhayu.ghosh17@gmail.com, swapnilpowerju@gmail.com, ndjana.cse@nitdgp.ac.in

Abstract—Audio-visual speech synthesis (AVSS) is an emerging research topic in the paradigm of generative AI, aiming to generate realistic and synchronized audio-visual outputs for a target speaker based on input audio from any source speaker, combining Voice Conversion (VC) and Audio-Visual Synthesis (AVS). The VC component transforms the audio characteristics of the source speaker to match the target speaker, while the AVS component generates a video that is synchronized with the converted audio. This paper introduces a novel approach to AVSS by incorporating the Kolmogorov Arnold Network (KAN) into the discriminator of Generative Adversarial Networks (GANs) for both VC and AVS tasks. By integrating KAN into the GAN framework, the proposed approach leverages KAN's ability to model high-dimensional, non-linear relationships with fewer parameters, enhancing the discriminator's efficiency and accuracy. This results in more natural synthesis of audio-visual outputs with reduced computational complexity and faster training time. Experimental evaluations on benchmark datasets demonstrate that our KAN-based GAN substantially outperforms existing methods in the quality and naturalness of generated samples. This work advances the state of AVSS by improving the realism of synthesized speech and video, offering a robust and efficient solution for creating synchronized audio-visual output.

Index Terms—Audio-Visual Speech Synthesis, Voice Conversion, Audio-Visual Synthesis, Generative Adversarial Networks, Kolmogorov-Arnold Networks.

I. INTRODUCTION

Audio-Visual Speech Synthesis (AVSS) is an emerging field in AI and multimedia processing that generates synchronized audio-visual outputs, such as talking head videos, from audio inputs [1]. Its goal is to produce realistic videos of the target speaker with precise audio-visual alignment, particularly in lip movements. AVSS has applications in virtual avatars, human-computer interaction, film dubbing, and accessibility tools [2]. It involves two key tasks: Voice Conversion (VC) [3], which adapts a source speaker's voice to match a target speaker, and Audio-Visual Synthesis (AVS) [4], which generates synchronized visual outputs. The process of creating a target speaker's audio-visual stream from the source speaker's input audio is known as AVSS [5], [6].

Despite advancements, current AVSS methods struggle to produce highly realistic and synchronized outputs. Traditional deep learning models, often with complex architectures and many parameters, suffer from computational inefficiencies and scalability issues [7], [8]. They also struggle to capture the

non-linear relationships between audio and visual data, resulting in unnatural or poorly synchronized outputs, especially when the source and target speakers differ significantly in style or appearance [9]. Although GANs [10], [11] have been used to enhance realism, their discriminators often fail to fully exploit audio-visual correlations, limiting the output quality.

To address these challenges, we propose integrating the Kolmogorov Arnold Network (KAN) [12] into the GAN framework for AVSS, termed KANGAN-AVSS. KAN is used in the discriminators of both VC and AVS modules, leveraging its ability to model complex, high-dimensional relationships with fewer parameters. This improves the discriminator's efficiency [13], reducing computational costs, speeding up training, and enhancing scalability without compromising output quality. The KAN-based discriminator also strengthens the ability to differentiate between real and generated content, resulting in more realistic and synchronized audio-visual outputs. Furthermore, its reduced complexity enhances the stability and convergence of GAN training [14], making KANGAN-AVSS a robust, efficient solution for high-quality AVSS.

We validated the KANGAN-AVSS model through extensive experiments on the VoxCeleb2 [15] and LRS-TED [16] datasets. Objective metrics like Mel-Cepstral Distortion (MCD) [17], Modulation Spectra Distance (MSD) [18], and F0-RMSE [19], as well as subjective Mean Opinion Scores (MOS) [20], were used for comprehensive evaluation [21]. The results demonstrate that KANGAN-AVSS outperforms state-of-the-art methods in terms of both audio fidelity and visual synchronization.

The primary contributions of this research can be summarized as follows:

- We propose an innovative AVSS approach that integrates the KAN into the discriminator of GAN model, enhancing the model's ability to capture complex audio-visual correlations and produce highly realistic outputs.
- The proposed approach demonstrates that the use of KAN enables the construction of a more efficient model with fewer parameters, resulting in reduced computational complexity and faster training times without compromising on the quality of the synthesized content.
- The integration of KAN into the discriminator significantly improves its ability to distinguish between real and

synthesized audio-visual data, leading to more accurate and natural synthesis.

- We validate our approach on two challenging datasets, demonstrating the robustness and generalizability of the proposed model across a wide range of speaker characteristics and real-world scenarios, and the experimental results show that the proposed KANGAN-AVSS outperforms existing methods in terms of evaluation metrics.
- This work advances the state-of-the-art (SOTA) in AVSS by addressing key limitations in current methods, offering a more efficient, accurate, and scalable solution for generating synchronized audio-visual content that is indistinguishable from real-life recordings.

This paper is organized as follows: Section II describes the proposed methodology, while Section III details the experimental setups of this work. Section IV presents and discusses the results, and Section V concludes the paper.

II. PROPOSED METHODOLOGY

In this section, we present the detailed methodology of the proposed KANGAN-AVSS, leveraging the unique capabilities of KAN to enhance the discriminator's effectiveness in GAN, leading to more realistic audio-visual outputs. Furthermore, Fig. 1 depicts the overall model architecture of the proposed KANGAN-AVSS framework.

A. Voice Conversion (VC) Module

The VC module takes as input the audio signal from a source speaker and transforms it to match the vocal characteristics of a target speaker [3]. Let, $x_s(t)$ denote the audio signal of the source speaker at time t , and $y_t(t)$ denote the corresponding audio signal of the target speaker. The goal of the VC module is to learn a transformation function f_{VC} such that:

$$y_t(t) = f_{VC}(x_s(t), e_t), \quad (1)$$

where, e_t is the embedding vector representing the target speaker's identity.

We employ a VC generator G_{VC} for this purpose, that processes the input audio signal $x_s(t)$ to generate the transformed audio signal $y_t(t)$ while being conditioned on the target speaker embedding e_t . Moreover, the attention mechanism [22] is applied to ensure that the generator focuses on the relevant parts of the input signal during synthesis, enhancing the accuracy of the transformation. Furthermore, VC discriminator plays the role to differentiate the transformed audio signal whether it is real or fake, which is later discussed in Section II-C.

B. Audio-Visual Synthesis (AVS) Module

The AVS generator G_{AVS} generates a synchronized video sequence $V(t)$ that corresponds to the converted audio signal $y_t(t)$. It can be mathematically represented as:

$$V(t) = G_{AVS}(y_t(t)). \quad (2)$$

This generator consists of a visual feature extractor and a temporal dynamics model. We use a convolutional neural

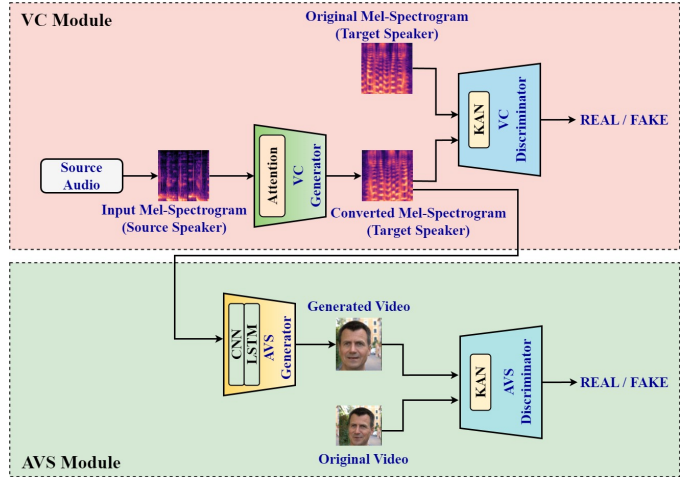


Fig. 1. Overview of the proposed KANGAN-AVSS framework.

network (CNN) [23] to extract spatial features from the target speaker's facial images. Let, $I(t)$ denote the input image frame at time t . The CNN extracts a feature map $F(t)$ as follows:

$$F(t) = \text{CNN}(I(t)). \quad (3)$$

To model the temporal dynamics of facial movements and lip synchronization, we employ a recurrent neural network (RNN) [24], such as LSTM [25]. The RNN takes as input the extracted feature map $F(t)$ and the converted audio signal $y_t(t)$ and generates the synthesized video frame $V(t)$:

$$V(t) = \text{RNN}(F(t), y_t(t)). \quad (4)$$

The RNN captures the sequential dependencies between video frames, ensuring that the synthesized video is temporally coherent and synchronized with the audio. Furthermore, the AVS discriminator works on synthesized samples by the generator, whether they are real or fake in nature.

C. KAN-based Discriminators

The discriminator plays a crucial role in the GAN framework by distinguishing between real and synthesized samples [26]. In this work, we integrate the KAN into the discriminators of the VC and AVS modules, enhancing its ability to model these intricate relationships with fewer parameters.

KAN is based on the Kolmogorov-Arnold superposition theorem [27], which states that any multivariate continuous function can be represented as a superposition of continuous univariate functions and a summation. In the context of our discriminator, KAN is used to decompose the high-dimensional input space into a series of simpler univariate functions, which are then combined to form the final decision function [28].

Let, $y_t(t)$ and $V(t)$ represent the synthesized audio and video signals, respectively. The discriminators D_{KAN}^{VC} and D_{KAN}^{AVS} output probabilities p_{VC} and p_{AVS} indicating whether they are real or synthesized:

$$p_{VC} = D_{KAN}^{VC}(y_t(t)), \quad (5)$$

$$p_{AVS} = D_{KAN}^{AVS}(V(t)). \quad (6)$$

The input signals of the discriminators are first processed by a series of univariate functions ϕ_i , each representing a specific aspect of the data. For VC and AVS models, it can be represented as follows:

$$\phi_i^{VC} = \phi_i([y_t(t)]), \quad (7)$$

$$\phi_i^{AVS} = \phi_i(V(t)). \quad (8)$$

The outputs of these univariate functions are combined using a summation layer, which aggregates the information from different functions.

$$\sum_{i=1}^N \phi_i^{VC} = \phi_1^{VC}, \phi_2^{VC}, \dots, \phi_N^{VC}, \quad (9)$$

$$\sum_{i=1}^N \phi_i^{AVS} = \phi_1^{AVS}, \phi_2^{AVS}, \dots, \phi_N^{AVS}. \quad (10)$$

The aggregated sum is passed through a final decision layer to produce the output probabilities p_{VC} and p_{AVS} , respectively:

$$p_{VC} = \sigma\left(\sum_{i=1}^N \phi_i^{VC}\right), \quad (11)$$

$$p_{AVS} = \sigma\left(\sum_{i=1}^N \phi_i^{AVS}\right), \quad (12)$$

where, $\sigma(\cdot)$ is the sigmoid activation function [29].

The KAN-based discriminator's ability to efficiently capture the intricate correlations between input features results in a more accurate and robust adversarial training process.

D. Training Objective and Loss Functions

The training of our KAN-based GAN model is driven by adversarial learning [30], where the generator and discriminator are optimized simultaneously for both the sequential processes (VC and AVS). The generator aims to minimize the adversarial loss, while the discriminator aims to maximize it. The adversarial loss for the VC model is defined as follows:

$$\mathcal{L}_{VC} = \mathbb{E}_{(y_t)} [\log D_{KAN}^{VC}(y_t)] + \mathbb{E}_{\hat{y}_t} [\log(1 - D_{KAN}^{VC}(\hat{y}_t))], \quad (13)$$

where, y_t and $\hat{y}_t$ represent the real and the generated audio features, respectively.

Similarly, the adversarial loss for AVS model will be:

$$\mathcal{L}_{AVS} = \mathbb{E}_{(V)} [\log D_{KAN}^{AVS}(V)] + \mathbb{E}_{\hat{V}} [\log(1 - D_{KAN}^{AVS}(\hat{V}))], \quad (14)$$

where, V and $\hat{V}$ are the real and the generated video frames, respectively.

To ensure that the synthesized outputs closely match the target audio-visual content, we introduce a reconstruction loss, as shown in Eq. 15:

$$\mathcal{L}_{recon} = \mathbb{E} [\|y_t - \hat{y}_t\|_2^2] + \mathbb{E} [\|V - \hat{V}\|_2^2], \quad (15)$$

This loss encourages the generator to produce such outputs that are perceptually similar to the real data. Therefore, the total loss for the generator is a weighted sum of the adversarial and reconstruction losses:

$$\mathcal{L}_{total} = \mathcal{L}_{VC} + \mathcal{L}_{AVS} + \lambda \mathcal{L}_{recon}, \quad (16)$$

where λ is a hyperparameter that balances the two types of losses. Moreover, The discriminator is trained to maximize the adversarial loss, forcing the generator to improve the realism and synchronization of the synthesized audio-visual content.

III. EXPERIMENTAL SETUPS

A. Dataset Descriptions

For training and evaluation, we utilize the VoxCeleb2 [15] and LRS-TED [16] datasets, both of which are widely recognized for their high-quality audio-visual content. VoxCeleb2 contains over 1 million utterances from more than 6,000 speakers, offering diverse speaker characteristics and challenging scenarios for VC and synchronization tasks. LRS-TED, on the other hand, provides a rich dataset of TED talks with accurate lip synchronization and facial expressions. These datasets together ensure that our model is exposed to a wide range of real-world conditions, enhancing its ability to generalize and perform effectively across different speakers and contexts.

B. Model Training and Implementation Details

The KANGAN-AVSSS is trained on the VoxCeleb2 and LRS-TED datasets, optimized using the RAdam optimizer [34]. Training begins with a small learning rate, which decreases gradually to ensure stable convergence. A batch size of 32 is chosen to balance efficiency and gradient stability. The model's performance is evaluated using objective metrics like MCD [17], MSD [18], and F0-RMSE [19], and subjective metrics like MOS [20], ensuring the generation of high-quality, synchronized audio-visual content. The source code of the proposed KANGAN-AVSS is publicly shared at: <https://github.com/Subhayu-ghosh/KANGAN-AVSS>.

IV. RESULTS AND DISCUSSIONS

This section presents the experimental results of the proposed KANGAN-AVSS and compares its performance with SOTA models. The VC performance is evaluated using MCD, MSD, F0-RMSE, and MOS. It is noteworthy that, the lower MCD, MSD, and F0-RMSE values, and the higher MOS values indicate superior performance [35] in transforming the source speaker's voice to resemble the target speaker's characteristics, while maintaining high-quality and natural-sounding audio. The performance of the KANGAN-AVSS is compared against various SOTA models in literature.

Table I compares the performance of the proposed KANGAN-AVSS model against several SOTA models, including StarGAN-VC2 [31], RNCapsGAN-VC [32], Exemplar-AE [9], ViTAE-AVSS [6], and CCLCapAE-AVSS [33], across two datasets. On the VoxCeleb2 dataset, KANGAN-AVSS achieves the lowest MCD of 5.24, MSD of 1.51, and F0-RMSE of 0.54,

TABLE I
VC RESULTS COMPARISON OF KAN-BASED GAN WITH SOTA MODELS ON VOXCELEB2 AND LRS3-TED DATASETS

Model	VoxCeleb2 Dataset				LRS3-TED Dataset			
	MCD ($\downarrow$)	MSD ($\downarrow$)	F0-RMSE ($\downarrow$)	MOS ($\uparrow$)	MCD ($\downarrow$)	MSD ($\downarrow$)	F0-RMSE ($\downarrow$)	MOS ($\uparrow$)
StarGAN-VC2 [31]	5.98	1.82	0.64	2.89	6.11	1.96	0.66	2.48
RNCapsGAN-VC [32]	5.61	1.70	0.59	3.28	5.75	1.77	0.63	2.92
Exemplar-AE [9]	5.65	1.71	0.59	3.10	5.77	1.81	0.64	2.87
ViTAE-AVSS [6]	5.41	1.62	0.57	3.72	5.69	1.74	0.60	3.45
CCLCapAE-AVSS [33]	5.30	1.54	0.55	3.98	5.66	1.69	0.58	3.74
KANGAN-AVSS (Ours)	5.24	1.51	0.54	4.04	5.48	1.55	0.56	3.83

TABLE II
AVS RESULTS COMPARISON OF KAN-BASED GAN WITH SOTA MODELS ON VOXCELEB2 AND LRS3-TED DATASETS

Model	MOS (VoxCeleb2)	MOS (LRS3-TED)
LipNet [36]	2.74	2.68
Wav2Lip [4]	2.82	2.75
Exemplar-AE [9]	2.83	2.91
ViTAE-AVSS [6]	3.84	3.72
CCLCapAE-AVSS [33]	4.02	3.82
KANGAN-AVSS (Ours)	4.11	3.96

along with the highest MOS of 4.04, indicating superior VC performance and perceptual quality. Similarly, on the LRS3-TED dataset, KANGAN-AVSS outperforms other models with MCD, MSD, F0-RMSE, and MOS values of 5.48, 1.55, 0.56, and 3.83, respectively, demonstrating its robustness across datasets.

Table II presents a comparative evaluation of the proposed KANGAN-AVSS model against several SOTA models, including LipNet [36], Wav2Lip [4], Exemplar-AE [9], ViTAE-AVSS [6], and CCLCapAE-AVSS [33], using the MOS as the primary evaluation metric. On the VoxCeleb2 dataset, the KANGAN-AVSS model achieves the highest MOS of 4.11, surpassing all other models. The next best performance is by CCLCapAE-AVSS with a MOS of 4.02. Similarly, on the LRS3-TED dataset, KANGAN-AVSS maintains its leading position with a MOS of 3.96. Although slightly lower than its performance on the VoxCeleb2 dataset, it still outperforms the other models, with CCLCapAE-AVSS achieving a MOS of 3.82, signifying the superiority of the proposed approach.

Furthermore, the consistent improvement across datasets underscores the model's reliability, making it a strong candidate for applications in virtual avatars, dubbing, and multimedia processing. The experimental results clearly show that the proposed KAN-based GAN model significantly outperforms the compared SOTA models in both VC and AVS tasks. The improvements in MCD, MSD, F0-RMSE, and MOS metrics highlight the effectiveness of the KAN in capturing complex audio-visual correlations with fewer parameters, leading to more efficient and accurate synthesis. Additionally, the subjective evaluations of generated videos suggest that the outputs generated by our model are more natural and better synchronized, enhancing the overall user experience.

To further assess the effectiveness of the key components in our KANGAN-AVSS model, we conducted an ablation study by systematically removing or modifying specific elements

TABLE III
ABLATION STUDY OF KANGAN-AVSS ON VOXCELEB2 DATASET

Model Variation	MOS (VC)	MOS (AVS)
Baseline GAN (without KAN)	3.56	2.92
KAN with Increased Parameters	4.02	4.05
KANGAN-AVSS (Full Model)	4.04	4.11

and evaluating the resulting performance, as presented in Table III. We replaced the KAN in the discriminator with a standard CNN to observe the impact on performance. The results showed a significant drop in MOS scores across both datasets, confirming the critical role of KAN in improving the discriminator's ability to model complex audio-visual correlations. Then, We experimented with a version of KAN that utilized a higher number of parameters, where the overall MOS scores remained comparable in both VC and AVS, indicating that the efficiency of KAN with fewer parameters is sufficient for high-quality synthesis.

In summary, The superior performance of the proposed KANGAN-AVSS can be attributed to its ability to model high-dimensional, non-linear relationships more efficiently, resulting in better generalization and robustness across diverse speaker characteristics and scenarios. These results underscore the potential of our approach in advancing the state of AVSS technology, offering a scalable and efficient solution for generating high-quality, synchronized audio-visual content.

V. CONCLUSION

The proposed KANGAN-AVSS model significantly advances the field of AVSS by integrating the KAN into the GAN framework. The experimental results demonstrate that KANGAN-AVSS consistently outperforms SOTA models in both VC and AVS tasks across multiple datasets. The incorporation of KAN enhances the model's ability to capture complex audio-visual relationships with fewer parameters, leading to high-quality, natural-sounding outputs. This efficiency and effectiveness make KANGAN-AVSS a robust solution for real-world applications, such as virtual avatars, dubbing, and multimedia processing. Future work will focus on optimizing the KANGAN-AVSS model for improved performance and scalability, while enhancing its generalization across diverse speakers and environments, and extending its applications to real-time systems and multilingual AVSS.

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